LITTLE STREAM DETECTOR

LSD 3.0 Technical Documentation

Property	Value
Release	3.0
Edition	Standard and fragmented MP4 support
Runtime	Windows PowerShell 5.1 + .NET Framework
Architecture	Single PowerShell file with embedded C#
External multimedia tools	None
Technical status	Functional / PASS
Revision	9 September 2026

Current architecture. Conventional MP4 and fragmented MP4 use separate sample-indexing branches. Both create the same canonical sample model and reuse the existing codec, audio, bitrate, tooltip, report, and A/B layers. The fragmented MP4 route has been validated with a real AVC/AAC asset, including complete video, audio, keyframe, frame-type and SliceQPY accounting.

1. Scope and Design Goals

Little Stream Detector 3.0 is a native Windows media inspection application. PowerShell provides orchestration and the WinForms interface, while embedded C# performs binary parsing, canonical sample construction, frame-header traversal, validation, and statistic accumulation.

- Analyze supported containers and codecs without launching FFmpeg, FFprobe, MediaInfo, or another multimedia executable.
- Keep container addressing separate from codec interpretation.
- Read only bounded samples that pass file-range validation.
- Report unavailable information as N/A instead of estimating values.
- Preserve stable routes while adding new functionality as isolated branches.

Supported families

Area	Families
Containers	Matroska, WebM, AVI 1.0, OpenDML AVI 2.0, MP4, M4V, MOV
Video	AVC, HEVC, AV1, VP8, VP9, MPEG-4 Part 2 / XviD
Audio	AAC, MP3, PCM, AC-3, E-AC-3
Outputs	Summary A/B, Streams, JSON, Log, bitrate profile and quantizer distribution

2. Architecture and Processing Pipeline

File selection -> container detection -> sample indexing -> canonical media model -> codec/audio analysis -> completed result -> reports and A/B graphs

Layer	Responsibility
Presentation	WinForms window, reports, graphs and interactive bitrate tooltip
Coordination	Background work, cancellation, progress and result promotion
Container parsing	Native tracks, timing, offsets and sample sizes
Canonical adaptation	Container-neutral bounded samples
Codec analysis	AVC, HEVC, AV1, VP8, VP9 and MPEG-4 Visual
Comparison	Immutable Reference B snapshot and compatible A/B overlays

MP4 branch selection

The conventional MP4 branch is used when classic sample tables provide active samples. The fragmented MP4 branch is used when media samples are described by movie fragments. The conventional path remains separate and unchanged.

3. Canonical Media Model

Object	Purpose
LsdMediaModel	Source, container, duration, timecode scale and tracks
LsdTrackModel	Track ID/type, codec family, codec-private data, payload format and samples
LsdSampleModel	File offset, length, DTS, PTS, duration and key state
LsdCodecPacket	Container-neutral packet passed to a codec analyzer

Both MP4 branches produce LsdSampleModel records. Codec analyzers do not read MP4 boxes and require no fMP4-specific parsing logic.

Boundary contract

- FileOffset is non-negative and Length is positive.
- FileOffset + Length remains inside the source file.
- Fragment-derived ranges must resolve to media payload data.
- PayloadFormat must match the selected codec analyzer.

4. Container Support

Container	Sample indexing route
Matroska / WebM	EBML tracks, blocks, lacing and timestamps
AVI / OpenDML	RIFF chunks, idx1 and OpenDML indexes
Standard MP4 / MOV	stsd, stsz, stsc, stco/co64, stts, ctts and stss
Fragmented MP4	mvex/trex and moof/traf/tfhd/tfdt/trun

Fragmented MP4 support is a container-indexing feature. Once canonical samples are available, existing video and audio analyzers run unchanged.

5. Fragmented MP4 Sample Indexing

A fragmented file normally retains initialization metadata in moov while media samples are described by repeated moof and mdat pairs.

Box	Role
mvex / trex	Per-track default description index, duration, size and flags
moof / traf	Fragment and track grouping
tfhd	Track ID, base-data offset and fragment defaults
tfdt	Base media decode time, version 0 or 1
trun	Sample count, data offset, duration, size, flags and composition offset
mdat	Resolved media payload ranges

Inheritance order

trun sample field -> tfhd fragment default -> trex track default

Resolved values become FileOffset, Length, DTS, PTS, Duration and IsKeyFrame in the canonical sample model.

6. Video Analysis Modules

Family	Native analysis
H.264 / AVC	SPS/PPS/VUI, I/P/B and SliceQPY
H.265 / HEVC	VPS/SPS/PPS, I/P/B and SliceQPY
AV1	OBU traversal, multi-tile headers and Base Q Index
VP8	Frame tag, boolean-coded first partition, KEY/INTER and Base Q Index
VP9	Superframes, uncompressed header, references, shown/hidden and Base Q Index
MPEG-4 Part 2 / XviD	VOL/VOP, I/P/B/S, GMC and VOP quantizer

The codec parser selected by the canonical codec family is shared across supported containers, including the fragmented MP4 route.

7. Audio Analysis

Family	Validation and reporting
AAC	AudioSpecificConfig, core/output rate, channels, SBR, PS, payload and bitrate
MP3	Version/layer, bitrate, sample rate, padding and frame size
AC-3 / E-AC-3	Syncword, bitstream ID, rate, frame size, blocks and channels
PCM	Block alignment, sample format, duration and bitrate

Fragmented audio samples are converted to the same canonical audio list and processed by the existing CanonicalAudioAnalyzer.

8. Bitrate and Quantizer Analysis

Every canonical video sample contributes its size to presentation-time bitrate bins. The mouse-hover tooltip displays the selected interval and the available A/B bitrate values. The final bucket ends exactly at the active duration, preventing a zero-length interval and an artificial bitrate spike.

Codec	Histogram value
AVC / HEVC	Frame-level SliceQPY
AV1	Base Q Index
VP8	Base Q Index
VP9	Base Q Index
MPEG-4 Part 2	VOP quantizer

9. GUI and A/B Comparison

View	Content
Summary A / B	Independent readable current and reference reports
Streams	Structured native track information
JSON	Serializable metadata and streams
Log	Lifecycle, validation and first-failure messages
Bitrate profile	Blue A and gold B series with hover tooltip
Quantizer panel	Mode-aware A/B statistics and distributions

- A Current is the most recently analyzed result.
- B Reference is an immutable stored comparison result.
- Exactly one A and one B are supported.
- The fMP4 branch adds no new GUI state.
- Quantizer distributions are directly comparable only when A and B use the same quantizer metric.

10. Validation and Defensive Parsing

- Track IDs must match initialized tracks.
- Sample duration, size and flags must resolve through trun, tfhd or trex.
- Computed ranges must remain inside the file.
- DTS is derived from tfdt and cumulative sample duration.
- PTS is derived from DTS and composition offsets.
- No guessed value is substituted when inheritance or offset resolution fails.
- Codec histograms remain disabled when parser completeness checks fail.
- A successful active sample model is not reported when both MP4 branches produce zero usable samples.

11. Support Matrix and Known Limitations

Video	Matroska/WebM	Standard MP4/MOV	fMP4	AVI
H.264 / AVC	Yes	Yes	Yes*	Yes, Annex B
H.265 / HEVC	Yes	Yes	Yes*	No
AV1	Yes	Yes	Yes*	No
VP8	Yes	No	No	No
VP9	Yes	Yes, vp09	Yes, vp09*	No
MPEG-4 Part 2	Yes	Yes	Yes*	Yes

* The fMP4 indexer reuses the existing codec parser when the fragment sample entry and canonical payload format are supported.

Known limitations

- The fragmented MP4 branch supports standard unencrypted files containing an initialization moov box and media fragments described by moof, traf, tfhd, tfdt and trun.
- Encrypted CENC samples and auxiliary encryption structures such as senc, saiz and saio are not supported.
- Standalone media segments without an initialization moov box are not supported.
- Incomplete files that are still being written and live network streams are not supported.
- VP8 is supported only in Matroska/WebM. VP8 in MP4/MOV and AVI is intentionally not enabled.
- VP9 in AVI is not supported.
- AVI H.264/AVC support is intended for Annex B streams.
- AVI files containing AAC or E-AC-3 audio are not currently supported.
- Container-only color metadata, such as MP4 colr/nclx or Matroska Colour, is not currently used as a fallback.
- Quantizer distributions are directly comparable only when A and B use the same quantizer metric.
- LSD does not decode pixels or calculate PSNR, SSIM or VMAF.

12. Regression and Maintenance Rules

The retained fragmented AVC/AAC MP4 can be compared with a conventional MP4 containing the same elementary streams. A separately re-encoded MP4 remains useful for route and parser regression, but its payload, bitrate and quantizer histogram are not expected to match the source.

Check	Expected relation
Video/audio sample count	Equal for a lossless remux, or fully explained by container edits
Payload bytes	Equivalent elementary-stream accounting for a lossless remux
Keyframe and I/P/B counts	Equal for unchanged elementary streams
SliceQPY count and histogram	Equal for unchanged AVC/HEVC elementary streams
First and last timestamps	Equivalent after documented edit-list effects
Rejected samples/headers	Zero for a validated asset

Validated fMP4 asset

A real AVC/AAC fragmented MP4 completed with 10,863 video samples, 10,863 parsed AVC frame headers, 10,863 SliceQPY values, zero rejected video headers, 18,716 canonical AAC samples and zero rejected audio samples. The conventional MP4 branch was also regression-tested after the fMP4 change and remained functional.

Stable-code rules

- Keep the conventional and fragmented MP4 indexers separate.
- Do not add fMP4-specific behavior to codec parsers.
- Keep fragment inheritance explicit and validated.
- Do not broaden support to encrypted or initialization-free segments without dedicated assets.
- Retain a stable LSD 3.0 baseline and inspect exact diffs.

Appendix A. Key Types

Type / state	Responsibility
NativeMp4Probe	Common MP4 inspection and branch selection
Mp4Trex	Per-track defaults from trex
Fragment sample list	Resolved fragment samples before canonical adaptation
Mp4ToLsdAdapter/2.0	Conventional MP4 canonical route
FragmentedMp4ToLsdAdapter/3.0	Fragment-derived canonical route
LsdSampleModel	Shared bounded payload and timing contract
Codec analyzers	Unchanged frame-level parsing after canonical adaptation
ReferenceResult	Frozen B report, duration, bitrate bins and quantizer data

Appendix B. Terminology

Term	Definition
Conventional MP4	MP4 whose active samples are described by moov sample tables
Fragmented MP4	MP4 whose media samples are described by movie fragments
trex	Per-track default sample values stored under mvex
tfhd	Track fragment header and fragment-level defaults
tfdt	Base media decode time for a track fragment
trun	Track run containing sample count and optional sample fields
Canonical sample	Container-neutral bounded sample with timing and key state

Release identification. Little Stream Detector remains version 3.0. Fragmented MP4 is implemented as an isolated sample-indexing route and uses the existing canonical analysis pipeline.